

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M.S. Dhivyadharsan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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Course: CSA 5116

Course Code: Cryptography

Date: 6/10/20

CO1 - AT2

Concept Mapping

3. Construct a Concept map for Classical Encryption schemes: Substitution Techniques and Transposition Techniques. Perform the following Tasks.
- Show differences and similarities
 - Map how each affects the data structure and Security Level.
 - Analyse which is more vulnerable to attack and why.

Ans:

Classical Encryption Schemes

Substitution Techniques

- Replace Characters with other characters
- Keeps the same order
- eg: Caesar cipher

Transposition Techniques

- Rearrange the position of characters
- Characters remain the same
- eg: Rail Fence cipher.

i) Effect on Data structure and Security Level

- Substitution : Changes Character Values, Keep order.
- Transposition : Keeps Characters, Change order.
- Both affect data structure and give low Security on Their own.

a) Differences & Similarities

Feature	Substitution	Transposition
What Changes	Characters	Positions
Data structure	Character Values Change	order Change
Security Level	Low	Low
eg:	Caesar cipher	Rail fence cipher.

b) Which is more Vulnerable and Why?

- Substitution → Vulnerable to frequency analysis
- Transposition → Vulnerable because The original Character are still present only The order is Changed.

4. Create a Concept map Connecting steganography Concepts : Data Hiding, Watermarking and Survivability . perform the following Tasks.
- Show relationships b/w These Concepts
 - Analyse how each contributes to Secure Communication.
 - Include one real - world application for each.

Ans:

Steganography

(hides information secretly)

Data hiding	Water marking	Survivability
→ Hides secret data inside a medium	→ Add ownership/ Identification	→ Keeps hidden data even after changes

→ Secret data + medium (image/audio/video)

↓

Hidden information

a) Relationships

- Data hiding → focuses on Secrecy
- Water marking → for ownership / authentication
- Survivability → Ensure The hidden Data.

b) Contribution to Secure Communication

- Data hiding → Hides the existence of the message
- Watermarking → proves authentically and ownership.
- Survivability → Keeps the message safe from changes even after modifications.

c) Real World Applications.

- Data hiding → Secret message in an image
- Water marking → proves authenticity and ownership.
- Survivability → watermark remains after JPEG Compressions.